

TruthLens

Product Requirements Document (PRD)

Version	1.0 — UNESCO Youth Hackathon 2026
Product Owner	Nada
Target Event	UNESCO Youth Hackathon 2026
Status	Draft — Ready for Dev Handoff

1. Project Overview

TruthLens is an interactive, gamified web platform that trains users to identify AI-generated (Deepfake) videos from real ones. Inspired by the Duolingo learning model, the platform rewards progress, builds digital literacy skills, and explains the visual cues that expose manipulated media.

Mission Statement

Empower the next generation to navigate a world saturated with AI-generated misinformation by building media literacy through engaging, science-backed training.

1.1 Problem Statement

By 2026, synthetic media (Deepfakes) will represent a significant portion of online video content. The vast majority of people cannot reliably distinguish between real and AI-generated video, creating vulnerability to misinformation, political manipulation, and fraud.

1.2 Solution

A progressive learning loop:

- Present a video (real or Deepfake)
- User makes a decision: Real or Fake?
- System reveals the truth + explains the detection clues
- User earns XP, levels up, and unlocks harder challenges
- Dashboard tracks progress and skill score over time

2. Target Users & Personas

Persona	Profile	Primary Goal
The Student	Age 16–25, social media native, consumes short-form video daily	Wants to impress peers, learn fast, earn badges
The Concerned Adult	Age 35–55, aware of misinformation but lacks technical skills	Wants a simple, trustworthy guide to verify content
The Educator	Teacher or NGO trainer looking for classroom tools	Needs shareable, structured content for group sessions

3. Scope

3.1 In Scope (MVP — Hackathon Deliverable)

- Video challenge rounds (Real vs. Fake quiz)
- Post-round explanation: detection clues, visual artifacts
- XP / scoring system with level progression
- User session state (no login required — browser session)
- Curated dataset of 20–30 video clips (real + deepfake)
- Responsive web app (desktop + mobile)
- Arabic + English UI support

3.2 Out of Scope (Post-Hackathon)

- User accounts / authentication
- Real-time AI detection engine
- Live social media scanning
- Video upload by users
- Leaderboards across users

4. User Stories

4.1 Onboarding

ID	User Story	Acceptance Criteria
US-01	As a new user, I want to understand what the app does before I start, so I know what to expect.	Landing page shows a 3-step explainer. CTA button is visible above the fold.
US-02	As a user, I want to choose my language (EN/AR) on first visit.	Language switcher visible on landing page. Choice persists in session.
US-03	As a user, I want to choose a difficulty level before starting.	Three options shown: Beginner / Intermediate / Advanced. Selection is stored in session.

4.2 Core Challenge Loop

ID	User Story	Acceptance Criteria
US-04	As a user, I want to watch a video clip and decide if it is real or fake.	Video plays auto (muted). Two buttons visible: "Real" and "Fake". No answer shown during video.
US-05	As a user, I want immediate feedback after my choice, so I know if I was right.	Correct: green overlay + XP animation. Wrong: red overlay. Both reveal the truth label.
US-06	As a user, I want to understand WHY a video is fake, so I can learn the detection clues.	Explanation card appears after every round. Includes: clue category, description, visual callouts on video frame.
US-07	As a user, I want to see my score update in real-time after each round.	Score counter animates after feedback. Accuracy % visible in header.
US-08	As a user, I want to proceed to the next challenge with one click.	"Next Challenge" button appears after explanation card. Keyboard shortcut: Space / Enter.

4.3 Progression & Gamification

ID	User Story	Acceptance Criteria
US-09	As a user, I want to see my level and XP bar so I feel motivated to continue.	Persistent XP bar in header. Level name displayed (e.g., "Novice Detector"). Bar animates on XP gain.
US-10	As a user, I want to unlock badges for completing milestones.	Badges awarded at: first 5 rounds, 80% accuracy streak, completing a difficulty tier. Badge gallery visible in profile.
US-11	As a user, I want a summary screen after finishing a set of 10 rounds.	End-of-set screen shows: score, accuracy, XP gained, badges unlocked, and a "Play Again" CTA.

4.4 Education Hub

ID	User Story	Acceptance Criteria
US-12	As a user, I want to read a guide on how Deepfakes are made, so I understand the technology.	"How it works" page accessible from nav. Covers: GAN basics, face-swapping techniques, common artifacts.
US-13	As a user, I want a quick-reference checklist of Deepfake detection signs.	Downloadable/shareable "5-Second Check" card. Available in EN and AR.

5. Functional Requirements

5.1 Video Dataset Requirements

- Minimum 20 video clips (10 real, 10 deepfake) for MVP
- Each clip: 5–15 seconds, MP4 format, max 10 MB per clip
- Each clip tagged with: label (real/fake), difficulty level, detection clues (array), category
- Dataset stored as JSON manifest: dataset.json
- Source: FaceForensics++ public dataset or equivalent open license

5.2 Detection Clue Categories

The system must support the following clue taxonomy:

- Facial Artifacts — unnatural blinking, skin texture, hairline irregularities
- Audio Sync — lip movement not matching speech, unnatural prosody
- Lighting Inconsistency — mismatched shadows, reflections in eyes
- Edge Blur — soft or flickering borders around face/hair
- Background Anomalies — warping, ghosting in background elements

5.3 Scoring Algorithm

Event	XP Awarded	Notes
Correct answer	+10 XP base	Base reward
Correct + fast (<5s)	+15 XP	Speed bonus
Correct on Advanced clip	+20 XP	Difficulty multiplier
Wrong answer	+0 XP	No penalty to avoid discouragement
3-round streak	+5 XP bonus	Streak multiplier

5.4 Level Progression

Level	Title	XP Required	Unlocks
1	Novice Detector	0 XP	Beginner clips only
2	Apprentice	100 XP	Intermediate clips
3	Analyst	300 XP	Advanced clips

4	Expert	600 XP	Expert badge + share card
5	Truth Guardian	1000 XP	Full dataset access + certificate

6. Non-Functional Requirements

Category	Requirement	Metric
Performance	Page must load in < 3 seconds on average connection	Lighthouse score ≥ 85
Video Load	Video clip must begin playing within 2 seconds	Lazy-load + preload next clip
Responsiveness	Full functionality on mobile (360px+) and desktop (1280px+)	No horizontal scroll on mobile
Accessibility	WCAG 2.1 AA compliance. Keyboard navigable.	All interactive elements have ARIA labels
Localization	Full Arabic (RTL) and English (LTR) support	Language switch requires no page reload
Privacy	No PII collected. Session data stays in localStorage only.	No backend calls to third-party analytics in MVP

7. Recommended Tech Stack

⚡ Developer Note

This stack is optimized for a Hackathon environment: fast setup, minimal configuration, and demo-ready in < 2 days.

Layer	Technology	Reason
Frontend Framework	Next.js 14 (React)	Fast routing, built-in i18n, Vercel deploy in 1 click
Styling	Tailwind CSS	Rapid UI iteration, RTL support via "dir" attribute

Animations	Framer Motion	XP bar animation, feedback overlays, badge reveals
State Management	Zustand	Lightweight, no boilerplate, perfect for session state
Video Playback	Native HTML5 <video>	No dependencies, full control over events
Data Layer	Static JSON + localStorage	Zero backend needed for MVP
Deployment	Vercel	Free tier, instant deploy from GitHub push

8. Pages & Routing

Route	Page Name	Key Components
/	Landing / Home	HeroSection, HowItWorks, StartButton, LanguagePicker
/setup	Difficulty Setup	DifficultyCard × 3, UserNameInput (optional)
/play	Challenge Screen	VideoPlayer, ChoiceButtons (Real/Fake), TimerBar, XPHeader
/result	Round Result	VerdictOverlay, ExplanationCard, ClueHighlighter, NextButton
/summary	Set Summary	ScoreCard, AccuracyChart, BadgeReveal, PlayAgainCTA
/learn	Education Hub	DeepfakeGuide, ClueGlossary, DetectionChecklist
/profile	User Progress	LevelBadge, XPProgressBar, BadgeGallery, StreakTracker

9. Data Models

9.1 Video Clip Object

Each entry in dataset.json follows this schema:

Field	Type	Description
id	string	Unique identifier, e.g. "clip_001"
label	"real" "fake"	Ground truth label
difficulty	"beginner" "intermediate" "advanced"	Difficulty tier
videoUrl	string	Path to MP4 file (relative or CDN URL)
thumbnailUrl	string	Preview frame for loading state
clues	string[]	Array of clue category keys (see §5.2)
explanation	{ en: string, ar: string }	Bilingual explanation text
source	string	Dataset attribution (e.g., "FaceForensics++")

9.2 User Session Object (localStorage)

Field	Type	Description
totalXP	number	Cumulative XP earned this session
level	number	Current level (1–5)
accuracy	number	Correct answers / total rounds (0–1)
currentStreak	number	Consecutive correct answers
badges	string[]	Array of earned badge IDs
difficulty	string	Selected difficulty for current set
language	"en" "ar"	UI language preference
completedClipIds	string[]	IDs of clips already shown

10. MVP Sprint Plan (Hackathon Timeline)

🕒 Assumption

Hackathon duration = 48 hours. Team = 2 developers + 1 designer. This plan assumes work begins immediately after kickoff.

Phase	Hours	Deliverables	Owner
Phase 0	0 – 4 hrs	Project setup, Vercel deploy, dataset.json structure	Lead Dev
Phase 1	4 – 16 hrs	Landing page, Setup screen, core Play loop (video + buttons)	Both Devs
Phase 2	16 – 28 hrs	Result screen + explanations, XP system, Level progression	Both Devs
Phase 3	28 – 36 hrs	Summary screen, Badge system, Education Hub, Arabic RTL	Both Devs
Phase 4	36 – 44 hrs	UI polish, animations (Framer Motion), mobile fixes	Dev + Designer
Phase 5	44 – 48 hrs	Final QA, demo script, pitch deck prep	Whole Team

11. Definition of Done

A feature is considered DONE when ALL of the following are true:

- Functionality matches the acceptance criteria in §4 (User Stories)
- Works on Chrome, Firefox, and Safari (latest versions)
- Responsive on 375px (iPhone SE) and 1280px (desktop)
- Arabic and English UI both render correctly (RTL/LTR)
- No console errors in production build
- Tested manually with at least 3 full rounds in each difficulty
- Deployed to Vercel and accessible via public URL

Demo Checklist (Pitch Day)

Live URL works on judge's device | Full round playable in < 2 min | At least 1 badge earnable in the demo | Arabic mode works | Score/XP visible at all times

12. Risks & Mitigations

Risk	Severity	Mitigation
Dataset licensing issues — Deepfake video rights	HIGH	Use FaceForensics++ (academic license) or generate simple synthetic examples using free tools pre-Hackathon
Video files too large — slow load times	MEDIUM	Compress to < 5 MB per clip using ffmpeg. Use CDN hosting (Cloudinary free tier).
Arabic RTL breaks layout	MEDIUM	Test RTL early (Phase 1). Use Tailwind "rtl:" variant. Avoid absolute positioning.
Time runs out before badge system is complete	LOW	Badge system is Phase 3 — core loop (Phases 0–2) is fully playable without it. Cut badges if needed.